ARROW 3 WEAPON SYSTEM

# Requirements Consistency Report

A3-INT-SSS-010 Rev A · Interceptor Segment Specification vs. the program baseline

| 11 | 5 | 5 | 1 |
|---|---|---|---|
| TOTAL FINDINGS | CRITICAL | MAJOR | MINOR |

**VERDICT** — **NOT READY FOR BASELINE — 5 critical and 5 major inconsistencies must be resolved before the document can enter the PDR baseline.**

| | |
|---|---|
| **Document under review** | A3-INT-SSS-010 Rev A — Interceptor Segment Specification |
| **Baseline checked** | 6 documents uploaded so far · 98 requirements indexed |
| **Generated** | 22 October 2026, 14:56 |
| **Produced by** | EDK custom agent “Arrow 3 Baseline Sentinel” (Hermes) — automated analysis, to be confirmed by the Interface Working Group |
| **Classification** | UNCLASSIFIED // FICTITIOUS DEMONSTRATION DATA |

## Blocking findings

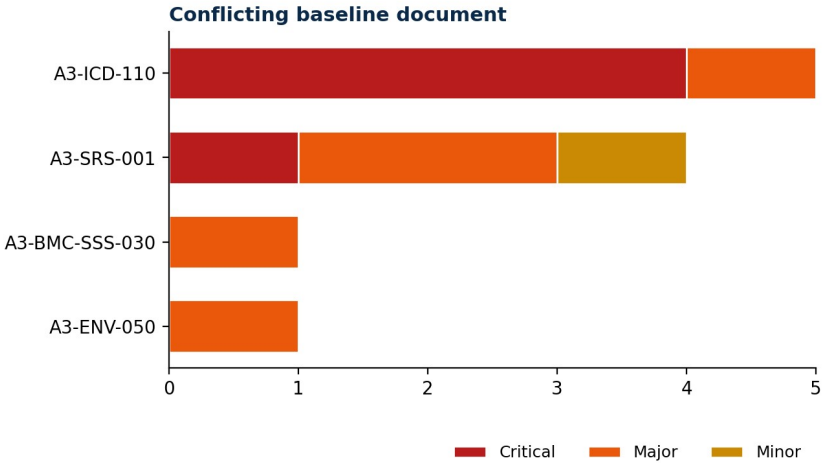
| | | |
|---|---|---|
| **F-01** | **Data Interface Mismatch** | INT-DAT-020 ↔ IF-DAT-001<br>A3-ICD-110 |
| **F-02** | **Coordinate System Mismatch** | INT-NAV-040 ↔ SRS-NAV-002<br>A3-SRS-001 |
| **F-03** | **Excessive All-Up Round Mass** | INT-PHY-002 ↔ IF-MEC-002<br>A3-ICD-110 |
| **F-04** | **Excessive Power Draw** | INT-PWR-012 ↔ IF-PWR-003<br>A3-ICD-110 |
| **F-05** | **Ready-to-Launch Timeline Overrun** | INT-TIM-031 ↔ IF-SEQ-006<br>A3-ICD-110 |

# 1. Executive summary

The incoming Interceptor Segment Specification (A3-INT-SSS-010 Rev A) was checked requirement by requirement against all 6 Arrow 3 documents uploaded so far. The check found 11 inconsistencies: 5 critical, 5 major and 1 minor. Most conflicts are with A3-ICD-110 (5).

Critical findings break a baselined interface and block the PDR baseline; major findings violate an allocated budget or environment; minor findings are documentation inconsistencies.




# 2. Conflict heat map

Where the incoming document disagrees with the baseline: each cell counts the findings for one category and one baseline document, coloured by the most severe of them.

| Category | BMC-SSS-030 | ENV-050 | ICD-110 | SRS-001 |
|---|---|---|---|---|
| Compliance | · | · | · | 1 |
| Electrical | · | · | 1 | · |
| Environmental | · | · | · | 1 |
| Interface | 1 | · | 1 | 1 |
| Performance | · | · | 1 | · |
| Physical | · | · | 1 | · |
| Reliability | · | 1 | · | 1 |
| Timeline | · | · | 1 | · |

Critical Major Minor

# 3. Findings at a glance

| ID | Severity | Finding | Incoming req. | Conflicts with |
|---|---|---|---|---|
| F-01 | CRITICAL | Data Interface Mismatch | INT-DAT-020 | IF-DAT-001<br>A3-ICD-110 |
| F-02 | CRITICAL | Coordinate System Mismatch | INT-NAV-040 | SRS-NAV-002<br>A3-SRS-001 |
| F-03 | CRITICAL | Excessive All-Up Round Mass | INT-PHY-002 | IF-MEC-002<br>A3-ICD-110 |
| F-04 | CRITICAL | Excessive Power Draw | INT-PWR-012 | IF-PWR-003<br>A3-ICD-110 |
| F-05 | CRITICAL | Ready-to-Launch Timeline Overrun | INT-TIM-031 | IF-SEQ-006<br>A3-ICD-110 |
| F-06 | MAJOR | Outdated IFTU Specification | INT-DAT-022 | BMC-ENG-020<br>A3-BMC-SSS-030 |
| F-07 | MAJOR | Reduced Operating Temperature Range | INT-ENV-050 | SRS-ENV-002<br>A3-SRS-001 |
| F-08 | MAJOR | Reliability Shortfall (MTBF) | INT-RAM-070 | RAM-004 |

| | | | | |
|---|---|---|---|---|
| | | | | A3-ENV-050 |
| **F-09** | **MAJOR** | Reduced Storage Life | INT-RAM-072 | SRS-RAM-010<br>A3-SRS-001 |
| **F-10** | **MAJOR** | Degraded Time Synchronization Accuracy | INT-TIM-035 | IF-TIM-010<br>A3-ICD-110 |
| **F-11** | **MINOR** | Outdated EMC Standard | INT-EMC-060 | SRS-EMC-001<br>A3-SRS-001 |

# 4. Detailed findings

## F-01 · Data Interface Mismatch

**CRITICAL** Interface · INT-DAT-020 ↔ IF-DAT-001 (A3-ICD-110)

| INCOMING · A3-INT-SSS-010 Rev A · INT-DAT-020 | BASELINE · A3-ICD-110 · IF-DAT-001 |
|---|---|
| *"The interceptor shall exchange pre-launch data with the launcher over a dual-redundant Gigabit Ethernet (1000BASE-T) link..."* | *"Data between the Launcher Control Unit and the AUR shall be exchanged over a dual-redundant MIL-STD-1553B bus..."* |

**Impact:** Complete physical and protocol incompatibility between Interceptor and Launcher.

**Recommended resolution:** Revert to MIL-STD-1553B or overhaul entire system data architecture.

## F-02 · Coordinate System Mismatch

**CRITICAL** Interface · INT-NAV-040 ↔ SRS-NAV-002 (A3-SRS-001)

| INCOMING · A3-INT-SSS-010 Rev A · INT-NAV-040 | BASELINE · A3-SRS-001 · SRS-NAV-002 |
|---|---|
| *"Pre-launch target data and in-flight target updates shall be received and processed in Earth-Centred Inertial coordinates (ECI, J2000 epoch)."* | *"All target state vectors and launch parameters exchanged between segments shall be expressed in WGS-84 Earth-Centred, Earth-Fixed (ECEF) coordinates..."* |

**Impact:** Targeting failure; the interceptor will process coordinates in a different frame than provided by the BMC and FCR.

**Recommended resolution:** Change interceptor processing to WGS-84 ECEF.

## F-03 · Excessive All-Up Round Mass

**CRITICAL** Physical · INT-PHY-002 ↔ IF-MEC-002 (A3-ICD-110)

| INCOMING · A3-INT-SSS-010 Rev A · INT-PHY-002 | BASELINE · A3-ICD-110 · IF-MEC-002 |
|---|---|
| *"The mass of the all-up round, including the canister, shall not exceed 1,450 kg."* | *"The mass of the loaded canister (all-up round) shall not exceed 1,350 kg."* |

**Impact:** Exceeds maximum mass allowed by the ICD and the launcher erector structural limits (LCH-MEC-014).

**Recommended resolution:** Reduce AUR mass to 1,350 kg or update Launcher and ICD specifications.

## F-04 · Excessive Power Draw

**CRITICAL** Electrical · INT-PWR-012 ↔ IF-PWR-003 (A3-ICD-110)

| INCOMING · A3-INT-SSS-010 Rev A · INT-PWR-012 | BASELINE · A3-ICD-110 · IF-PWR-003 |
|---|---|
| *"During pre-launch (IMU run-up and thermal conditioning) the interceptor shall draw not more than 18 A continuous from the launcher supply."* | *"The current drawn by one AUR from the launcher supply shall not exceed 12 A continuous in any pre-launch state..."* |

**Impact:** Will trigger electronic circuit breakers in the launcher PDU (LCH-PWR-021), preventing power-up.

**Recommended resolution:** Optimize power consumption to 12 A or upgrade Launcher PDU capacity.

## F-05 · Ready-to-Launch Timeline Overrun

**CRITICAL** Timeline · INT-TIM-031 ↔ IF-SEQ-006 (A3-ICD-110)

| INCOMING · A3-INT-SSS-010 Rev A · INT-TIM-031 | BASELINE · A3-ICD-110 · IF-SEQ-006 |
|---|---|

| | |
|---|---|
| *"The interceptor shall report READY within 6.0 s of receiving the Prepare-to-Launch (PTL) command."* | *"The interceptor shall report READY (message STS, bit 0) within 3.0 s of receiving PTL."* |

**Impact:** Violates the system reaction time budget (SRS-TL-002), potentially delaying the first launch beyond the required 8.0 s.

**Recommended resolution:** Reduce internal readiness sequence to 3.0 s.

## F-06 · Outdated IFTU Specification

| MAJOR | Interface · INT-DAT-022 ↔ BMC-ENG-020 (A3-BMC-SSS-030) |
|---|---|
| INCOMING · A3-INT-SSS-010 Rev A · INT-DAT-022 | BASELINE · A3-BMC-SSS-030 · BMC-ENG-020 |
| *"In-flight target update messages shall be decoded per A3-ICD-120 Rev A."* | *"The BMC shall transmit in-flight target updates (IFTU) ... per A3-ICD-120 Rev C."* |

**Impact:** Message decoding errors if Rev C contains breaking changes relative to Rev A.

**Recommended resolution:** Update reference to A3-ICD-120 Rev C.

## F-07 · Reduced Operating Temperature Range

| MAJOR | Environmental · INT-ENV-050 ↔ SRS-ENV-002 (A3-SRS-001) |
|---|---|
| INCOMING · A3-INT-SSS-010 Rev A · INT-ENV-050 | BASELINE · A3-SRS-001 · SRS-ENV-002 |
| *"The all-up round shall operate at ambient temperatures from −20 °C to +50 °C."* | *"All field-deployed equipment shall operate at ambient temperatures from −33 °C to +55 °C..."* |

**Impact:** Interceptor cannot be deployed in extreme cold/heat environments defined as required for the system.

**Recommended resolution:** Expand operating range to −33 °C to +55 °C.

## F-08 · Reliability Shortfall (MTBF)

| MAJOR | Reliability · INT-RAM-070 ↔ RAM-004 (A3-ENV-050) |
|---|---|
| INCOMING · A3-INT-SSS-010 Rev A · INT-RAM-070 | BASELINE · A3-ENV-050 · RAM-004 |
| *"The captive-carry MTBF of the all-up round, powered on the launcher, shall be at least 1,500 h."* | *"Interceptor (all-up round) captive-carry MTBF ... shall be at least 2,000 h."* |

**Impact:** Increases the frequency of interceptor failures and replacement needs, degrading system operational availability.

**Recommended resolution:** Increase MTBF to 2,000 h.

## F-09 · Reduced Storage Life

| MAJOR | Reliability · INT-RAM-072 ↔ SRS-RAM-010 (A3-SRS-001) |
|---|---|
| INCOMING · A3-INT-SSS-010 Rev A · INT-RAM-072 | BASELINE · A3-SRS-001 · SRS-RAM-010 |
| *"The interceptor shall have a storage life of at least 10 years in its sealed canister."* | *"The interceptor shall have a storage life of at least 15 years in its sealed canister..."* |

**Impact:** Significantly increases logistics costs and replenishment frequency for the interceptor stockpile.

**Recommended resolution:** Increase storage life to 15 years.

## F-10 · Degraded Time Synchronization Accuracy

| MAJOR | Performance · INT-TIM-035 ↔ IF-TIM-010 (A3-ICD-110) |
|---|---|
| INCOMING · A3-INT-SSS-010 Rev A · INT-TIM-035 | BASELINE · A3-ICD-110 · IF-TIM-010 |
| *"The interceptor time reference shall be synchronised to system time within ±10 µs."* | *"...the interceptor shall align its time reference within ±1 µs."* |

**Impact:** Reduced precision in time-tagging and synchronization, potentially affecting interceptor guidance accuracy.

**Recommended resolution:** Tighten synchronization requirement to ±1 µs.

### F-11 · Outdated EMC Standard

| MINOR | Compliance · INT-EMC-060 ↔ SRS-EMC-001 (A3-SRS-001) |
|---|---|
| **INCOMING · A3-INT-SSS-010 Rev A · INT-EMC-060** | **BASELINE · A3-SRS-001 · SRS-EMC-001** |
| *"The interceptor shall meet MIL-STD-461F requirements CE102, CS114, RE102 and RS103."* | *"All segments shall meet MIL-STD-461G, tailored per A3-ENV-050 Rev B."* |

**Impact:** Potential compliance gap as MIL-STD-461F is superseded by Rev G.

**Recommended resolution:** Update compliance requirement to MIL-STD-461G.

## 5. Action plan

Proposed actions for the next Interface Working Group and Configuration Control Board.

| ID | Action | Owner | Due |
|---|---|---|---|
| F-01 | Revert to MIL-STD-1553B or overhaul entire system data architecture. | Interface Working Group (INT/LCH) | Before PDR data package |
| F-02 | Change interceptor processing to WGS-84 ECEF. | Interface Working Group (INT/LCH) | Before PDR data package |
| F-03 | Reduce AUR mass to 1,350 kg or update Launcher and ICD specifications. | Interceptor IPT | Before PDR data package |
| F-04 | Optimize power consumption to 12 A or upgrade Launcher PDU capacity. | Interface Working Group (INT/LCH) | Before PDR data package |
| F-05 | Reduce internal readiness sequence to 3.0 s. | Weapon System Engineering | Before PDR data package |
| F-06 | Update reference to A3-ICD-120 Rev C. | Interface Working Group (INT/LCH) | PDR |
| F-07 | Expand operating range to −33 °C to +55 °C. | Specialty Engineering | PDR |
| F-08 | Increase MTBF to 2,000 h. | Specialty Engineering | PDR |
| F-09 | Increase storage life to 15 years. | Specialty Engineering | PDR |
| F-10 | Tighten synchronization requirement to ±1 µs. | Interceptor IPT | PDR |
| F-11 | Update compliance requirement to MIL-STD-461G. | Interceptor IPT | Next revision |

## Appendix A. Baseline documents checked

| Document | Rev | Title | Requirements |
|---|---|---|---|
| **A3-BMC-SSS-030** | A | Battle Management Center Specification | 11 |
| **A3-ENV-050** | B | Environmental and RAM Requirements | 15 |
| **A3-FCR-SSS-020** | B | Fire Control Radar Specification | 14 |
| **A3-ICD-110** | C | Interceptor–Launcher ICD | 19 |
| **A3-LCH-SSS-040** | B | Launcher Segment Specification | 21 |
| **A3-SRS-001** | C | System Requirements Specification | 18 |

## Appendix B. How this report was produced

1. Trigger: A3-INT-SSS-010 Rev A was dropped onto the agent in EDK, waking both Hermes agents.
2. Baseline register (main Hermes): index/arrow3-baseline.md holds every requirement of every other Word document uploaded so far; only new or changed documents are re-read.
3. Spec Reader (second Hermes, in parallel): converted the incoming document to Markdown with pandoc, extracted every "shall" statement and wrote a management briefing.
4. Main Hermes studied the register and the new requirement lines together, then compared every incoming requirement with the baseline.
5. This report was generated from the findings by the program report kit (consistency_report.py).

*Automated findings support, but do not replace, engineering review. Confirm each finding against the source documents before CCB submission.*